SENECA ENERGY HACKATHON 2026 · TEAM BRIEF

Tide

Only ever pay 3.9¢

WHAT WE'RE BUILDING

In Ontario, the same electricity costs about 4¢ at midnight and 39¢ at dinnertime. Tide is a $15 plug that waits for the cheap, clean hours — automatically. Plug in once, forget it.

FOR	UPDATED	STATUS	READ
The whole team	May 21, 2026	Viability proven	~6 minutes

Contents

What Tide is, why it wins, and how you can help — before we lock our idea.

SECTION I

I The 60-second version

If you read nothing else before we lock our idea, read this.

The hook

In Ontario, the *exact same* electricity costs about **4 cents at midnight and 39 cents at dinnertime**. Almost nobody notices — so they overpay every single day. **Tide is a small plug that quietly waits for the cheap hours and runs your stuff then.** You plug it in once and forget it.

Three things to know

- **What it is** — a ~$15 smart plug plus a simple app that runs your car charger, air conditioner, or appliances *only* when electricity is cheapest (and cleanest).
- **Why it's good** — it saves real money — hundreds of dollars a year — completely automatically, by using a discount Ontario already offers but almost nobody captures.
- **Why it wins** — at the demo, we put a real plug on the table and a light switches **on** the second the cheap hour hits. Every other team shows a chart on a screen. We show a thing that *does something*.

SECTION II

II Why this exists

Ontario built a great discount — then made it almost impossible to actually use.

The hidden 10× price

Ontario offers an electricity plan called **Ultra-Low Overnight**. Overnight it's dirt cheap. But in the late afternoon and evening — 4 to 9pm, when everyone's home cooking and running the AC — it jumps to almost **ten times more**, and nearly double the normal rate.

Time of day	Price per unit
Overnight (11pm–7am)	about 4¢ — the deal
Daytime & weekends	about 10–16¢
Dinnertime (weekdays 4–9pm)	about 39¢ — the trap

So the plan is a fantastic deal *only if you actually shift your big electricity use to the middle of the night.* And that's the problem: life gets in the way. You're not going to run the dishwasher at 2am or babysit your car charger. So people sign up, forget to shift, get hammered by the 39¢ evenings — and end up **paying more** than before.

> **THE OPPORTUNITY IN ONE LINE**
> The discount is real. Almost nobody captures it. Tide captures it for you, automatically — that gap is the entire project.

SECTION III

III What Tide is, and how it works

Think of a roommate who only ever charges the car at the cheapest minute of the night — and never forgets.

What it is

You don't change your life. You don't watch the clock. You plug your device into Tide, tell it “I need this done by 7am,” and walk away. It handles the rest.

How it works

1. **Get a small smart plug** (about $15) and plug your car charger, window AC, or dehumidifier into it.
2. **Tide checks two things every hour:** how *expensive* power is right now, and how *dirty* it is right now. Ontario publishes both, for free, every few minutes.
3. **It finds the best window tonight** — the cheapest, cleanest stretch that still finishes your job on time.
4. **It switches the plug on then, and off when done.** You did nothing.

> **WHAT'S A SMART PLUG?**
> A little block that sits between your wall outlet and your device. It can turn that device's power on and off on command — normally from a phone app, but in our case from Tide, automatically. They cost $15–25 at any electronics store.

SECTION IV

IV What it saves, and the moment that wins

Real dollars — and the ten seconds the judges remember.

What it saves (real numbers, not guesses)

If you shift a load off the pricey 4–9pm window to overnight:

What you plug in	Saves about
Electric car (full charge)	**~$1,760 / year**
Electric car (nightly top-up)	~$790 / year
Dehumidifier	~$320 / year
Window air conditioner	~$250 / year
Dishwasher	~$140 / year

The $15 plug pays for itself in **days** on an electric car, **weeks** on anything else. For one car charge: about **$2.20 down to $0.34 — roughly 85% off.**

The 10-second moment that wins the room

Picture the judges' table. We set down a real smart plug with a lamp (standing in for “your dishwasher”). The screen shows **6pm** — the priciest hour — and the word **WAIT**. Then we fast-forward to **11pm**, the cheap hour clicks in, and **the lamp turns on, right there in front of them.** A counter ticks up: money saved, pollution avoided.

Judges see 30 projects in an afternoon and remember *one* thing about each — if anything. Ours is “the one where the plug turned the light on by itself.” That's the whole strategy: be the team that brought something physical that *acted*, not another dashboard.

SECTION V

V Why it's real — and the honest risks

We built a working test version this week, with real Ontario data.

This is not a maybe

Confirmed by a working prototype:

- It pulls Ontario's live grid information correctly.
- It correctly decides to wait until late night to charge an electric car — cutting the cost by **85%**.
- The money math is correct and is automatically double-checked.
- There's already a working demo screen with the big **WAIT / GO NOW** display.

What that means for the team: the hard, risky, “will this even work” part is done. What's left is polish, design, and telling the story well — which is where the rest of us come in.

The honest risks (so nothing surprises us)

THE PLUG ON VENUE WI-FI

The plug needs to work on the venue's Wi-Fi. We chose a type that works without internet, and we'll always keep the on-screen version as a backup if the hardware acts up.

LEAD WITH MONEY, NOT THE PLANET

The “cleaner energy” angle is strongest in summer; in a deep winter cold snap, overnight power isn't *much* cleaner — but it's *always* about 10× cheaper. So we lead with the savings and mention the cleaner-energy benefit second.

SECTION VI

VI How you can help & what we decide together

Most of what's left needs no code at all.

Jobs up for grabs

- **Design & the look** — make the one screen and the slide deck beautiful (we have a color scheme started).
- **The pitch** — write and rehearse the 60-second pitch; lead with the “4¢ vs 39¢” hook.
- **The demo video** — film the lamp turning on; the first 5 seconds are the wow.
- **The story** — find a real Ontario family or renter the savings would matter to; names and specifics beat generic claims.
- **Buy & test the plug** — grab a ~$20 TP-Link Kasa plug so we can rehearse with real hardware.
- **Words** — make the README and submission page read well to a non-engineer judge.

What we should decide together

1. **The name.** “Tide” (power flows in and out like a tide) is the safe pick. A bolder option: just call it **“3.9”** — the cheap overnight rate, as the brand.
2. **Which appliance we show.** An EV charger has the biggest savings; a window AC or dehumidifier is more relatable. Leaning relatable.
3. **Money or planet first?** Recommendation: money first — it's what makes people act.

The two alternates (we pick one)

Tide is the front-runner, but for fairness:

- **Rooftop Roll Call** — a map naming Toronto's 100 best big rooftops that should have solar but don't (“if we built these, it'd power 31,000 homes”). Strong, but it points at potential rather than saving energy today.
- **Backwards Hour** — an app that turns saving energy into a daily “put your phone down” wellness ritual, timed to when the grid is dirtiest. Memorable, but harder to prove it works on stage.

Tide is the only one of the three that **saves real, measurable money and energy on the spot** — which is why it's the recommendation.

APPENDIX

§ Plain-English glossary

The only terms you'll hear us use.

Term	What it means
The grid	The province-wide electricity system everything plugs into.
kWh	The unit your hydro bill charges by — one “scoop” of electricity. We mostly talk in dollars instead.
Ultra-Low Overnight (ULO)	The Ontario plan with the cheap-overnight / pricey-evening split. The thing Tide is built around.
Smart plug	The $15 gadget that turns a device on and off on command.
Carbon intensity	How much pollution the electricity is causing right now. Overnight is cleaner (nuclear, water, wind); evenings burn more gas.
Gas peaker	A backup gas power plant Ontario switches on at busy evening hours — why dinnertime power is both pricier and dirtier.

Questions? Drop them in the team channel. Idea-lock is tomorrow — please skim and react so we go in aligned.